

Table 1

∇F	$(4x, 6y)$
H	$\text{diag}(4, 6)$
K	4
x^*	$(0, 0)$
α	$0 < \alpha < 0,125$
α^*	$0,1$
p	$0,6$

Table 2

k	$x^{(k)}$	$\nabla F(x^{(k)})$	$F(x^{(k)})$
0	$4, 2$	$(16, 62)$	64
1	$16/5, 2/5$	$64/5, 32/5$	$544/25$
2	256 $(256, 0, 0)$	$256/25, 32/25$	$8224/625$

Table 3

α	$x^0 + \alpha \cdot d$	F	right side	accept
1	$(-12, -30)$	7488	63,872	no
0,5	$(-4, -14)$	1600	63,936	no
0,25	$(0, -6)$	288	63,968	no
0,125	$(2, -2)$	40	63,984	yes

Hi GD owner, Team 703

Signed: ~~Amir~~ Amir Amanbay